

Molecular structure inference from molecular spectra

with transformers, eventually

Multimodal NMR → Molecular structure
based on the ~800'000 molecule synthetic dataset

[1] [Marvin Alberts et al.](#) “Unraveling Molecular Structure: A Multimodal Spectroscopic Dataset for Chemistry”. In: *NeurIPS 2024 Datasets and Benchmarks Track*. 2024. [url]

in turn based on

[2] [Daniel Lowe](#). *Chemical reactions from US patents (1976–Sep2016)*. figshare, 2017. [url]

and MestReNova software (NMR prediction plug-in)

GPT-2 reference implementation¹

Input: $x \in \mathbb{N}^{B \times T}$ (token indices)

Token embedding: $\mathbf{E}_{\text{tok}} = \text{Embedding}(x) \in \mathbb{R}^{B \times T \times d}$

Position embedding: $\mathbf{E}_{\text{pos}} = \text{Embedding}[1 : T]$

Combined: $\mathbf{H}^{(0)} = \text{Dropout}(\mathbf{E}_{\text{tok}} + \mathbf{E}_{\text{pos}})$

For each block $\ell = 1 \dots L$:

$$\mathbf{X} = \mathbf{H}^{(\ell-1)}$$

$$\mathbf{X} \leftarrow \mathbf{X} + \text{CausalSelfAttn}(\text{LayerNorm}(\mathbf{X}))$$

$$\mathbf{X} \leftarrow \mathbf{X} + \text{MLP}(\text{LayerNorm}(\mathbf{X}))$$

$$\mathbf{H}^{(\ell)} = \mathbf{X}$$

Final: $\mathbf{H}_{\text{final}} = \text{LayerNorm}(\mathbf{H}^{(L)})$

Logits: $\hat{y} = \mathbf{H}_{\text{final}} \cdot \mathbf{E}_{\text{tok}}^{\top}$ (weight tying)

¹Based on <https://github.com/karpathy/nanoGPT/blob/master/model.py>

Phase I – Pretraining

- ✓ Pretrain on SMILES inputs, and:
 - ✓ bias a hidden state towards ChemBERTa-v2,
 - ✓ hijack a hidden state to predict functional groups,
 - ✓ augment with non-canonical SMILES.
- Correctness self-assessment flag – after SMILES add ✓/✗
- Backtracking / inference-time reasoning – provide/teach *undo*
- Pretrain on the one-atom-one-peak “quasi-NMR”

Phase II – NMR to SMILES inference training

- ✓ More careful data split (correlations due to USPTO origin)
- Compare vs Alberts et al., 2025 [3]
- Compare H+C+HSQC vs H+HSQC

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Phase I – Pretraining

O=C([O-])[O-].[Ca+2][:][_][o][_][<=>]C([O-])([O-])=O.[Ca+2] ■

A given SMILES structure

Phase I – Pretraining

O=C([O-])[O-].[Ca+2][:][_][o][_][⇌]C([O-])([O-])=O.[Ca+2] ■

Unsupervised trigger token (molecular embedding)

Phase I – Pretraining

$$\text{O}=\text{C}([\text{O}^-])[\text{O}^-].[\text{Ca}+2][:] \text{[red box]} [\circ][\text{—}][\rightleftharpoons] \text{C}([\text{O}^-])([\text{O}^-])=\text{O}.[\text{Ca}+2] \blacksquare$$

Supervised hidden state (molecular embedding)

Phase I – Pretraining

$$\text{O}=\text{C}([\text{O}^-])[\text{O}^-].[\text{Ca}+2][:][][\text{O}][][\rightleftharpoons]\text{C}([\text{O}^-])([\text{O}^-])=\text{O}.[\text{Ca}+2] \blacksquare$$

Unsupervised trigger token (functional groups)

Phase I – Pretraining

$$\text{O}=\text{C}([\text{O}^-])[\text{O}^-].[\text{Ca}+2][:][][\circ][\text{---}][\rightleftharpoons]\text{C}([\text{O}^-])([\text{O}^-])=\text{O}.[\text{Ca}+2] \blacksquare$$

Supervised hidden state (functional groups)

Phase I – Pretraining

O=C([O-])[O-].[Ca+2][:][_][o][_][<=>]C([O-])([O-])=O.[Ca+2] █

Unsupervised trigger token (alt SMILES)

Phase I – Pretraining

O=C([O-])[O-].[Ca+2][:][_][o][_][<=>C([O-])([O-])=O.[Ca+2] ■

Distribution-supervised alternative SMILES

Phase I – Pretraining

$\text{O}=\text{C}([\text{O}^-])[\text{O}^-].[\text{Ca}+2][:] [_][\circ][_][\rightleftharpoons]\text{C}([\text{O}^-])([\text{O}^-])=\text{O}.[\text{Ca}+2]$ 

Supervised “end of sequence”

Phase I – Pretraining: losses definitions

- $\frac{1}{n}(\frac{1}{2}\|T\mathbf{e}\|_2^2 - \log \det T + \frac{n}{2} \log 2\pi)$ – embedding ($n = 384$)
- $\frac{1}{n}(\|T\mathbf{e}\|_1 - \log \det T + n \log 2)$ – functional groups ($n = 289$)
- KL divergence, position-averaged – next-token distribution
- Causal LM = $-\log p(\text{eos} \mid \text{prefix})$ – end-of-sequence token

combined à la Kendall et al. [4]:

$$\text{minimize} \quad \sum_t \left(\frac{1}{\sigma_t^2} \text{loss}_t + \log(\sigma_t^2) \right)$$

with a trainable σ_t for each task t .

Phase I – Pretraining: training setup

Run 1/3:

- 580'738 training samples, 72'189 validation samples (top PubChem)
- Max. SMILES length: 50 characters; “multimodal dataset” excluded
- 30 epochs on NVIDIA A10 (24GB) with batch size 64
- Optimizer `adabelief` with learning rate 10^{-4}
- No embedding or functional groups supervision
- One of five alt-SMILES per sample per epoch

`[code]` `[weights]`

Phase I – Pretraining: training setup

Run 2/3 (resumed from Run 1):

- 688'876 training samples, 85'873 validation samples (top PubChem)
- Max. SMILES length: 70 characters; “multimodal dataset” excluded
- 30 epochs on NVIDIA A10 (24GB) with batch size 64
- Optimizer `adabelief`, learning rate 10^{-4}
- With embedding and functional groups supervision
- One of five alt-SMILES per sample per epoch

`[code]` `[weights]`

Phase I – Pretraining: training setup

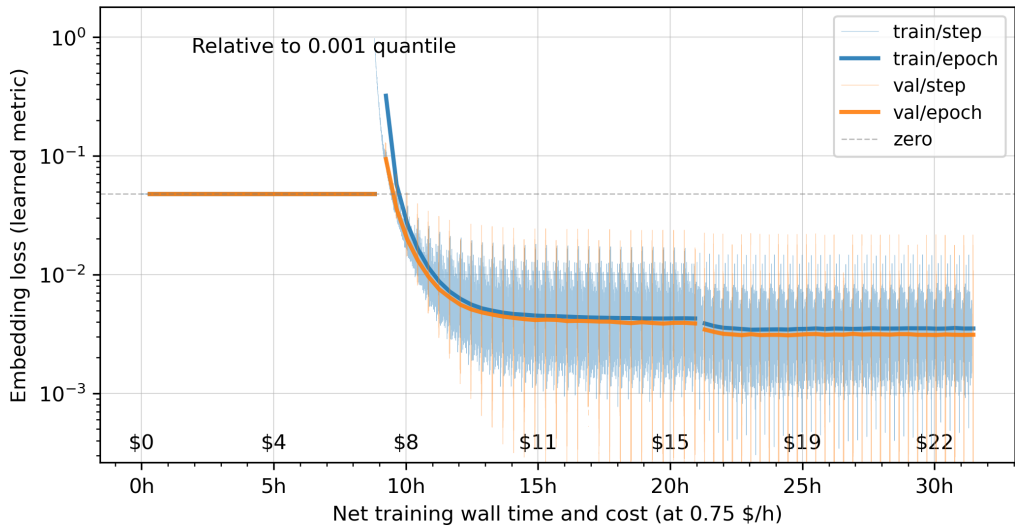
Run 3/3 (resumed from Run 2):

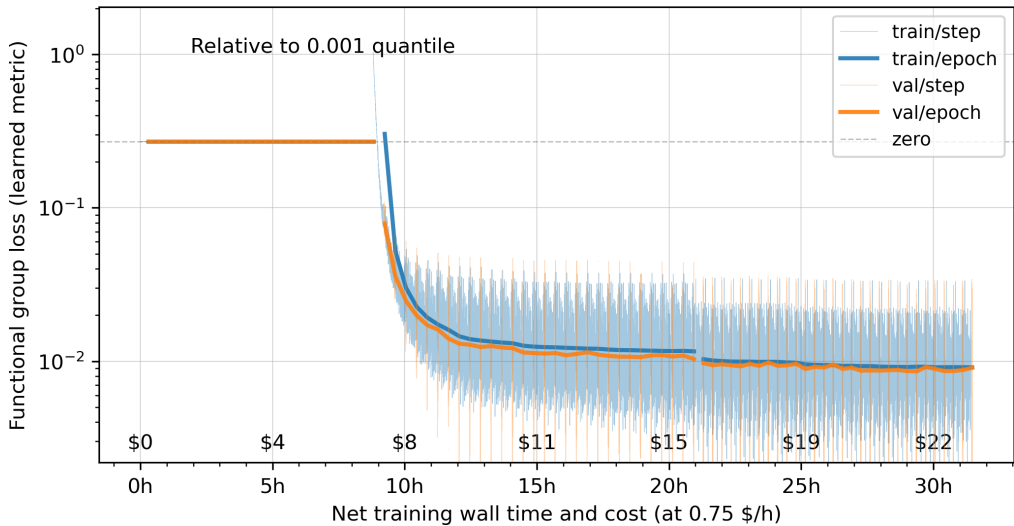
- 688'876 training samples, 85'873 validation samples (top PubChem)
- Max. SMILES length: 70 characters; “multimodal dataset” excluded
- 30 epochs on NVIDIA A10 (24GB) with batch size 96
- Optimizer `adabelief`, learning rate 10^{-4}
- With embedding and functional groups supervision
- One of five alt-SMILES per sample per epoch

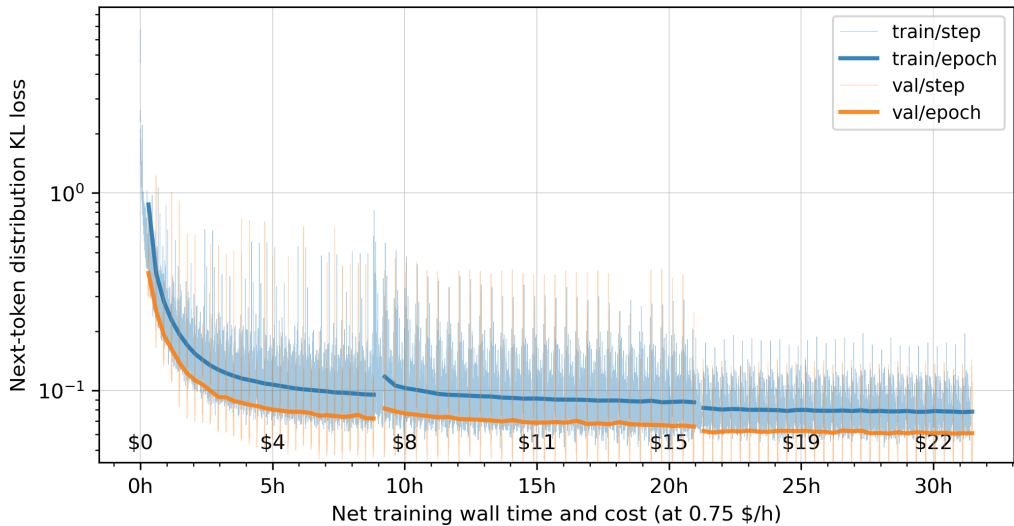
`[code]` `[weights]`

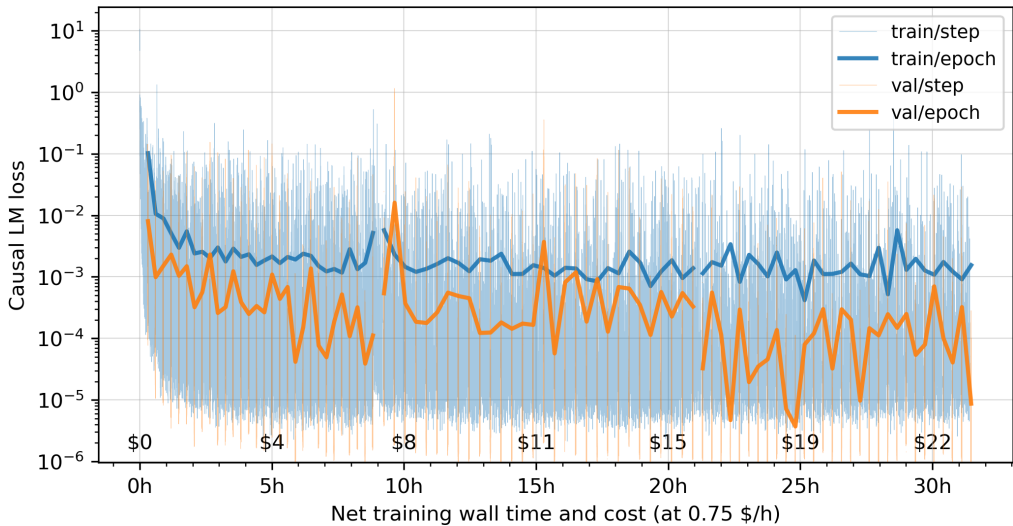
ONE
ETERNITY
LATER...

Losses
during training

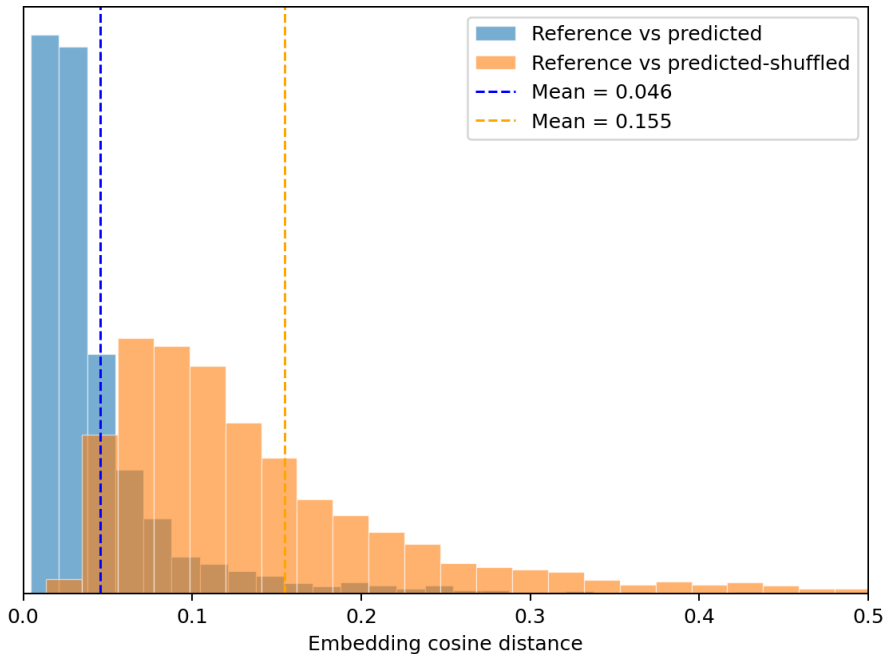






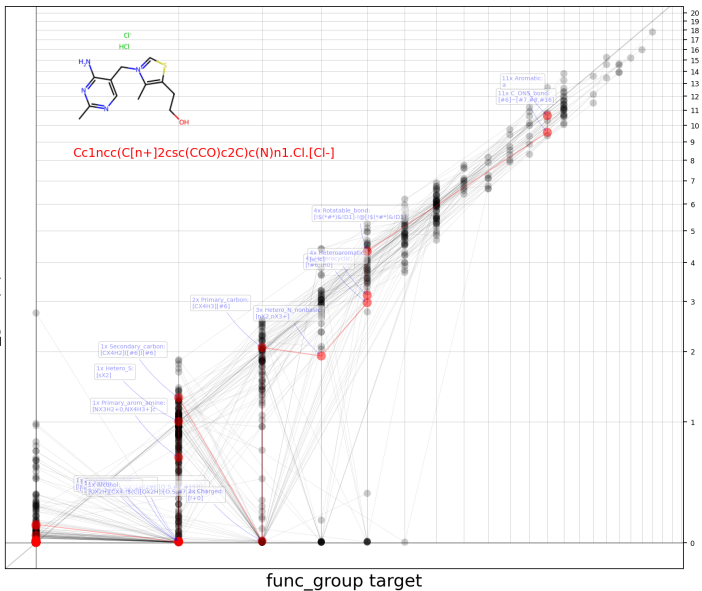


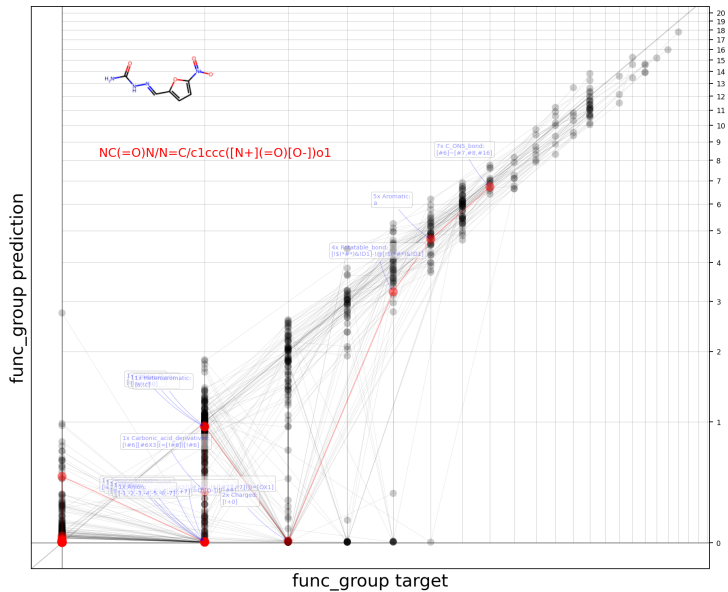
ChemBERTa-2 embedding
384-d vector



Functional groups
289-d vector of counts

func_group prediction





SMILES augmentation

supervision of next-token distribution

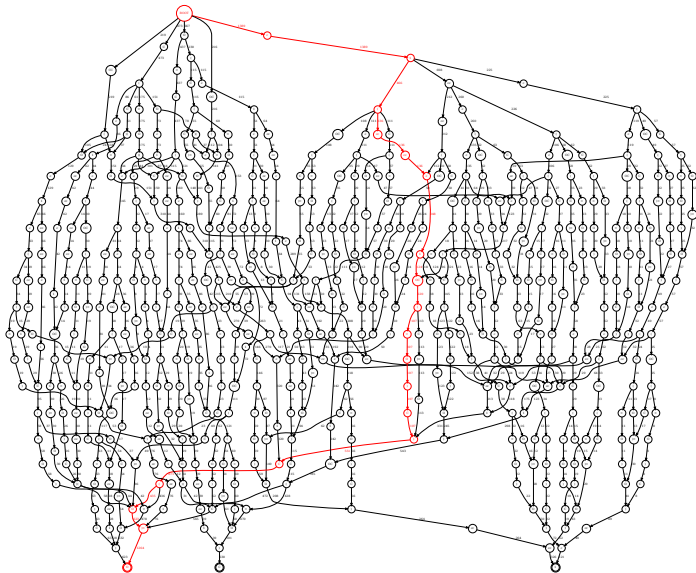
SMILES augmentation

Aspirin can be represented as:

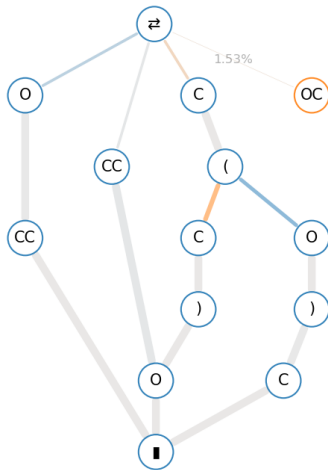
- O(c1ccccc1C(O)=O)C(=O)C
- c1(c(OC(C)=O)cccc1)C(=O)O
- c1cc(OC(C)=O)c(C(=O)O)cc1
- ...

304 variants total (unkekulized)

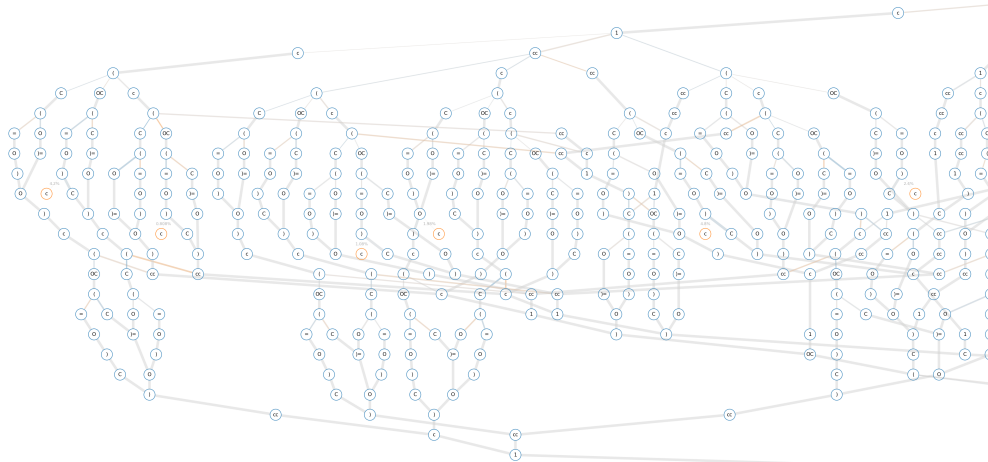
SMILES augmentation



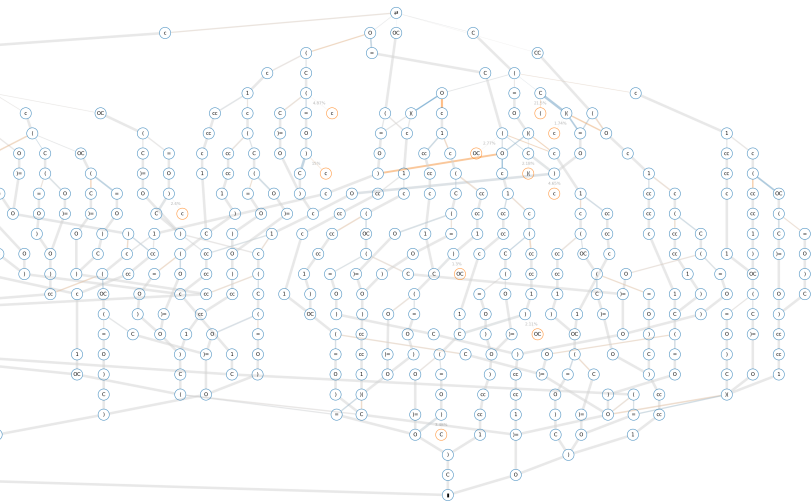
SMILES augmentation: ethanol



SMILES augmentation: aspirin (out-of-sample)



SMILES augmentation: aspirin (out-of-sample)



Are SMILES correct?
out-of-sample

Input SMILES: CC(C)(C)c1ccc(C(=O)NCc2ccco2)cc1

predicted variant	score	valid	correct	isomer
<chem>c1cc(C(C)(C)C)ccc1C(=O)NCc1occc1</chem>	-0.18	✓	✓	✓
<chem>c1c(ccc(c1)C(C)(C)C)C(=O)NCc1occc1</chem>	-0.18	✓	✓	✓
<chem>c1c(ccc(C(C)(C)C)c1)C(=O)NCc1occc1</chem>	-0.18	✓	✓	✓
<chem>c1cc(ccc1C(C)(C)C)C(=O)NCc1occc1</chem>	-0.18	✓	✓	✓
<chem>c1c(C(C)(C)C)ccc(c1)C(=O)NCc1occc1</chem>	-0.18	✓	✓	✓
<chem>c1cc(C(C)(C)C)ccc1C(NCc1occc1)=O</chem>	-0.19	✓	✓	✓
<chem>c1cc(C(C)(C)C)ccc1C(=O)NCc1ccco1</chem>	-0.19	✓	✓	✓
<chem>CC(c1ccc(cc1)C(=O)NCc1occc1)(C)C</chem>	-0.19	✓	✓	✓
<chem>c1c(ccc(C(C)(C)C)c1)C(=O)NCc1ccco1</chem>	-0.19	✓	✓	✓
<chem>c1c(ccc(c1)C(C)(C)C)C(=O)NCc1ccco1</chem>	-0.19	✓	✓	✓

Input SMILES: CC(C)C1CCC(Cc2ccc(Cl)cc2)C1(O)Cn1cncn1

predicted variant	score	valid	correct	isomer
<chem>c1cc(ccc1Cl)CC1CCC(C1(O)Cn1cncn1)C(C)C</chem>	-0.18	✓	✓	✓
<chem>c1cc(ccc1CC1CCC(C(C)C)C1(Cn1cncn1)O)Cl</chem>	-0.19	✓	✓	✓
<chem>c1cc(Cl)ccc1CC1CCC(C(C)C)C1(Cn1cncn1)O</chem>	-0.19	✓	✓	✓
<chem>c1cc(ccc1Cl)CC1CCC(C(C)C)C1(Cn1cncn1)O</chem>	-0.2	✓	✓	✓
<chem>c1cc(ccc1Cl)CC1CCC(C(C)C)C1(O)Cn1cncn1</chem>	-0.2	✓	✓	✓
<chem>c1cc(Cl)ccc1CC1CCC(C(C)C)C1(O)Cn1cncn1</chem>	-0.2	✓	✓	✓
<chem>CC(C)C1CCC(Cc2ccc(cc2)Cl)C1(O)Cn1cncn1</chem>	-0.2	✓	✓	✓
<chem>CC(C)C1CCC(Cc2ccc(cc2)Cl)C1(Cn1cncn1)O</chem>	-0.2	✓	✓	✓
<chem>c1c(ccc(c1)Cl)CC1CCC(C(C)C)C1(Cn1cncn1)O</chem>	-0.2	✓	✓	✓
<chem>c1cc(ccc1Cl)CC1CCC(C(C)C)C1(O)Cn1ncnc1</chem>	-0.22	✓	✓	✓

Input SMILES: COC(=O)C1=CO[C@@H](C)[C@@H]2CN3CCc4c([nH]c5cc(OC)ccc45)[C@@H]3C[C@H]12

predicted variant	score	valid	correct	isomer
<chem>O=C(OC)C1=CO[C@@H]([C@@H]2CN3CCc4c5ccc(OC)cc5[nH]c4[C@@H]3C[C@@H]21)C</chem>	-0.24	✓	✗	✓
<chem>O=C(OC)C1=CO[C@@H]([C@@H]2CN3CCc4c5ccc(OC)cc5[nH]c4[C@@H]3C[C@H]21)C</chem>	-0.25	✓	✗	✓
<chem>O=C(OC)C1=CO[C@@H]([C@@H]2CN3CCc4c5ccc(OC)cc5[nH]c4[C@H]3C[C@@H]21)C</chem>	-0.25	✓	✗	✓
<chem>O=C(OC)C1=CO[C@@H]([C@@H]2CN3CCc4c5ccc(OC)cc5[nH]c4[C@@H]3C[C@@H]12)C</chem>	-0.25	✓	✗	✓
<chem>O=C(OC)C1=CO[C@@H]([C@@H]2CN3CCc4c5ccc(OC)cc5[nH]c4[C@@H]3C[C@H]12)C</chem>	-0.25	✓	✗	✓
<chem>O=C(OC)C1=CO[C@@H]([C@@H]2CN3CCc4c5ccc(OC)cc5[nH]c4[C@H]3C[C@H]21)C</chem>	-0.25	✓	✗	✓
<chem>O=C(OC)C1=CO[C@@H]([C@@H]2CN3CCc4c5ccc(OC)cc5[nH]c4[C@H]3C[C@@H]12)C</chem>	-0.25	✓	✗	✓
<chem>O=C(OC)C1=CO[C@@H]([C@@H]2CN3CCc4c5ccc(OC)cc5[nH]c4[C@H]3C[C@H]12)C</chem>	-0.25	✓	✗	✓
<chem>O=C(OC)C1=CO[C@@H]([C@@H]2CN3CCc4c5ccc(OC)cc5[nH]c4[C@@H]2C1)C</chem>	-0.26	✗	✗	✗
<chem>O=C(OC)C1=CO[C@@H]([C@@H]2CN3CCc4c5ccc(OC)cc5[nH]c4[C@H]2C1)C</chem>	-0.26	✗	✗	✗

Input SMILES: N#N

predicted variant	score	valid	correct	isomer
N#N	-0.038	✓	✓	✓
C(#N)N	-0.3	✓	✗	✗
c1(ccccn1)N	-0.74	✓	✗	✗
C(#N)CN	-0.87	✓	✗	✗
c1c(N)#N	-0.95	✗	✗	✗
c1(ccccn1)#N	-0.97	✗	✗	✗
c1(cccc)N	-1	✗	✗	✗
C(N)#N	-1.1	✓	✗	✗
c1(N)cc	-1.1	✗	✗	✗
C(#N)CC	-1.1	✓	✗	✗

Paper summary: Alberts et al., 2025 [3]

[3] Marvin Alberts et al. *Automated structure elucidation at human-level accuracy via a multimodal multitask language model*. 2025. [url]

“Top–1 accuracies of up to 96%” on the “Chemical Education dataset”

[5] Scott E. Van Bramer and Loyd D. Bastin. “Spectroscopy data for undergraduate teaching”. In: *Journal of Chemical Education* 100.10 (2023), pp. 3897–3902. [url]

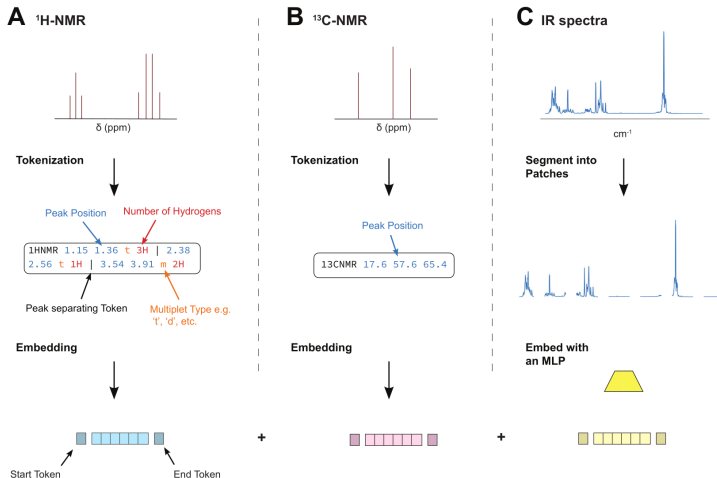


Figure 2: Combining different spectroscopic modalities. A and B) Tokenisation procedure for $^1\text{H-NMR}$ and $^{13}\text{C-NMR}$ spectra respectively. C) IR spectra are segmented into patches, and each patch is projected into the embedding space via an MLP. Bottom: The embeddings of each modality are flanked with a modality start and modality end token and concatenated. The concatenation of all modalities is fed into the model.

Step 1 – Synthetic pre-training

- 6-layer encoder–decoder transformer
- pretrained on 790'000 paired IR + C/H NMR spectra from [1]

Step 2 – Multitask architecture

- A single model handles every modality and their combinations
- Each modality block is wrapped in *modality-start* / *modality-end* tokens

Step 3 – Baseline finetune (171 paired spectra)

- Five-fold CV on the Chemical Education dataset
- Multimodal top-1: 69.10 ± 11.16 ; top-5: 91.47 ± 5.58

Modality	Multi	Simulated		Zero-shot		Finetuned	
		Top-1	Top-5	Top-1	Top-5	Top-1	Top-5
IR	$\times$	39.72	57.28	1.05	4.21	17.97 ± 5.65	44.37 ± 6.62
C	$\times$	49.35	66.85	48.42	66.32	57.75 ± 11.26	85.97 ± 9.62
H	$\times$	60.08	76.74	20.00	31.58	45.83 ± 8.14	72.89 ± 4.21
IR+C+H	$\times$	73.40	87.83	17.89	33.68	50.04 ± 11.25	81.26 ± 9.08
IR	$\checkmark$	33.57	53.21	2.11	11.58	15.97 ± 8.65	47.14 ± 8.30
C	$\checkmark$	41.37	61.80	53.16	82.11	58.57 ± 9.71	89.61 ± 4.69
H	$\checkmark$	53.44	72.82	31.58	51.58	49.96 ± 5.99	79.22 ± 3.91
IR+C+H	$\checkmark$	73.05	88.69	50.53	61.05	69.10 ± 11.16	91.47 ± 5.58

Table: Table 1 from Alberts et al., 2025 [3]

Step 4 – +600 unpaired spectra

- Added 600 molecules with only IR and/or ^{13}C data
- IR: NIST EPA gas-phase library; C-NMR: NMR-ShiftDB2
- Multimodal Top-1 rises to 75.41 ± 8.32

Step 4' – +33'088 unpaired spectra

- Further augmentation: $29'960 \times \text{C}$, $5'027 \times \text{IR}$, $1'899$ dual-modality
- IR: NIST EPA gas-phase library; C-NMR: NMR-ShiftDB2
- Multimodal Top-1 rises to 96.25 ± 7.50

Modality	Finetuned		+600 Samples		+33,088 Samples	
	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5
IR	15.97 $\pm$ 8.65	47.14 $\pm$ 8.30	21.69 $\pm$ 6.41	52.86 $\pm$ 9.33	42.50 $\pm$ 12.12	75.00 $\pm$ 6.85
C	58.57 $\pm$ 9.71	89.61 $\pm$ 4.69	63.55 $\pm$ 5.84	90.55 $\pm$ 5.37	96.25 $\pm$ 7.50	97.50 $\pm$ 5.00
H	49.96 $\pm$ 5.99	79.22 $\pm$ 3.91	52.77 $\pm$ 3.66	78.23 $\pm$ 7.27	93.75 $\pm$ 12.50	96.75 $\pm$ 2.50
IR+C+H	69.10 $\pm$ 11.16	91.47 $\pm$ 5.58	75.41 $\pm$ 8.32	91.52 $\pm$ 1.87	96.25 $\pm$ 7.50	98.75 $\pm$ 2.50

Table: Table 2 from Alberts et al., 2025 [3]

thx.

References I

- [1] Marvin Alberts, Oliver Schilter, Federico Zipoli, Nina Hartrampf, and Teodoro Laino. “Unraveling Molecular Structure: A Multimodal Spectroscopic Dataset for Chemistry”. In: *NeurIPS 2024 Datasets and Benchmarks Track*. 2024. [url].
- [2] Daniel Lowe. *Chemical reactions from US patents (1976–Sep2016)*. figshare, 2017. [url].
- [3] Marvin Alberts, Nina Hartrampf, and Teodoro Laino. *Automated structure elucidation at human-level accuracy via a multimodal multitask language model*. 2025. [url].
- [4] Alex Kendall, Yarin Gal, and Roberto Cipolla. “Multi-Task Learning Using Uncertainty to Weigh Losses for Scene Geometry and Semantics”. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (2018). [url].

References II

- [5] Scott E. Van Bramer and Loyd D. Bastin. “Spectroscopy data for undergraduate teaching”. In: *Journal of Chemical Education* 100.10 (2023), pp. 3897–3902. [url].

